

15. A farmer wishes to fence in a rectangular plot of ground bounded on one side by a river. He has 2200 feet of chain-link fence available. He will **not** fence the side along the river, but each of the two widths of the field perpendicular to the river will need a double layer (or “two-ply”) of fence. If the area of the rectangular plot enclosed is to be a maximum, find the number of feet in the length of the field that is parallel to the river. (Consider only the lengths of fencing used in the problem. Ignore the thickness of the double fencing in your computations.)
16. Find the absolute value in the difference of the two values of x for which the three terms $x - 2$, $x + 1$, and $4x - 8$ (in that order) will form a geometric progression?
17. (Multiple Choice) Write A if the conditions below determine a conic. Write B if the conditions below overdetermine a conic (that is, there is no conic meeting all the given conditions). Write C if the conditions below underdetermine a conic (that is, there is more than one conic that meets all the given conditions).

Ellipse with center at $(0, 0)$ and eccentricity of $\frac{2}{3}$

Note: Be certain to write the correct capital letter as your answer.

18. If $\sin(3x^\circ) = \frac{1}{2}$ and $133^\circ < x < 178^\circ$, find the value of x .
19. The length of the latus rectum of a hyperbola is 24.12, the eccentricity of the hyperbola is 3.62, the center of the hyperbola is at $(8.16, 19.02)$, and the principal axis of the hyperbola is parallel to the y -axis. One of the foci of the hyperbola is located at (x, y) . Find the largest possible value of y . Express your answer as a **decimal** rounded to the nearest hundredth.
20. Find the **absolute** value of the distance from the point $(1, 4, 9)$ to the plane determined by the 3 points $(5, 0, 4)$, $(1, 7, 3)$, and $(2, 8, 0)$. Express your answer as a **decimal rounded to the nearest thousandth**.